Supplementary material for Local order parameters for ice polymorphs: a unified comparison and practical guide

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S1 Beyond ice: key and confined water

The order parameters of the main text are often used to single out functionally important water in biomolecular simulations. As a transfer test outside the benchmark's design, we applied them unchanged, computed from water oxygens and hydrogens only (host, guest, protein and ions are not neighbors), to two sets of ligand-unbinding paths generated with PathGennie, using geometric definitions of the waters of interest that no order parameter enters. Scores are held-out balanced accuracies as in the main text: a threshold is fitted on one half of the paths (OAMe–G2: path number modulo 2) or on two replicas (T4L L99A: r1 and r2 against r3 and r4) and applied to the other, and the worse direction is reported. Frames are saved path states, not equally spaced in time. Table S1 summarizes the data.

Table S1: Data of the supplementary applications.

System	Paths	Model	Analyzed
OAMe–G2	208 PathGennie escape paths of guest G2	TIP3P, 300 K	up to 300 states per path
T4L L99A	89 benzene-egress paths: 48 with a dry cavity (r1, r4), 41 with three cavity waters (r2, r3)	OPLS-AA, SPC/E, 300 K	60 states per path

Host–guest unbinding (OAMe–G2). After rotating every frame onto the host template used by the path-sampling bias, a cavity water is one within 4.5 Å of the host axis and below the portal, and a bridging water one within 3.5 Å of both a guest polar atom and a host oxygen. The cavity is mostly dry. Averaged over the ten frames before the guest crosses the portal it holds no more than 0.5 waters in 185 paths, at least two in 8 and between in 6; 9 paths never cross. It stays nearly empty within the recorded path even after the guest has left, so dry and wet refer to occupancy during exit only. Against bulk water, cavity waters are separated by Voronoi density (0.99), smooth CN (0.91), $\bar{q}_6$ (0.85) and CN (0.87), and against water near the portal, which also touches the host, still by Voronoi density (0.91), smooth CN (0.85) and $\bar{q}_6$ (0.81; Fig. S1a). Bridging waters are as distinct from bulk (1.00, 0.91 and, through d_5, 0.84). Cavity waters of dry and wet paths are not distinguishable from each other (at most 0.64), and tetrahedral, entropy and ring parameters are unstable: $\bar{Q}$ separates cavity from portal water with an oracle score of 0.94 (threshold and test on all paths) but only 0.67 when the threshold

moves between path halves. Which waters will enter the cavity within the next ten saved states is decided by position: the distance to the portal alone gives a ROC area of 0.996, a held-out position model 0.997, and adding any or all order parameters does not improve it (Fig. S1b).

Buried water in a protein (T4 lysozyme L99A). In the wet ensemble three waters sit beside benzene in the hydrophobic cavity from the start. Thresholds fitted on two replicas and tested on the other two separate these three waters from protein-surface water through Voronoi density (0.99), smooth CN (0.97), $\bar{q}_6$ and CN (0.96) and $\bar{s}$ (0.89), but not through $\bar{Q}$ or d_5 (below 0.5). Other buried waters (at least 60 protein heavy atoms within 8 Å) are harder (at most 0.80), because many belong to small connected internal clusters. A rule fitted on buried versus surface water of one replica pair and applied to the other pair (mean of both directions) flags 3.4 waters per frame within 8 Å of the cavity in the wet ensemble, against three present, recovering 99.9% of their appearances, and 1.2 in the dry ensemble, mostly other buried waters near the pocket (Fig. S1c). Because the three cavity waters are the same molecules in both wet replicas and each replica pair holds one wet replica, these held-out tests rest on one wet trajectory set on each side. The three cavity waters remain buried and isolated (smooth CN about 1.7) after benzene has left.

Interpretation and limits. In both systems the parameters detect confinement through the water coordination deficit, the missing water neighbors that a hydrophobic cavity or a protein interior imposes, rather than through a distinct local structure: isolated waters have large $\bar{q}_6$ and q_3 only because a Steinhardt invariant of one or two bonds is large. Detection is reliable for isolated confined waters and weak for water in connected internal clusters, and local structure does not anticipate which waters enter a cavity. For confined water we would therefore use Voronoi density, smooth CN or CN from water neighbors only, and avoid tetrahedral, entropy and ring parameters and any reading of a high $\bar{q}_l$ of isolated waters as order. These are two systems, two water models at 300 K and sampled path frames rather than equilibrium trajectories; the results show what water-only parameters can and cannot report, not a general water classifier.

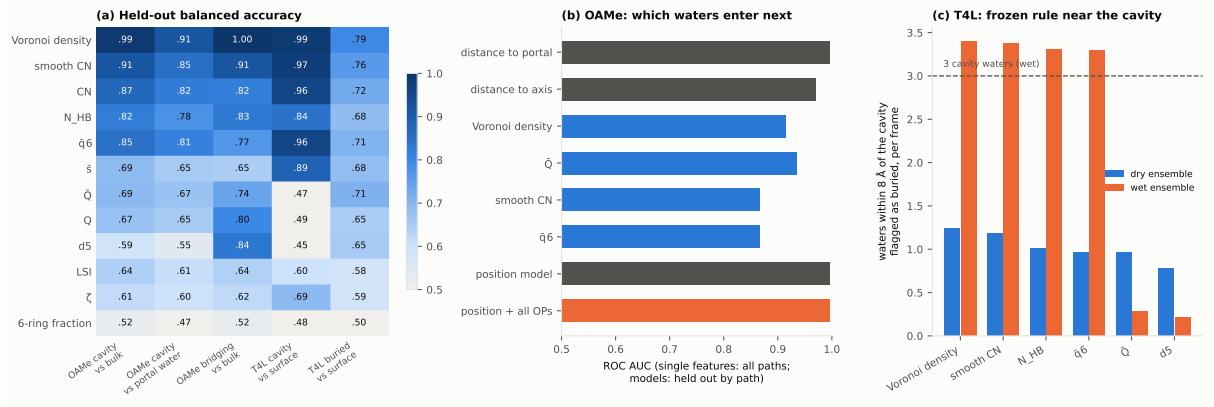


Figure S1: Key and confined water beyond ice. (a) Held-out balanced accuracy (threshold fitted on half of the paths or replicas, applied to the other, worse direction) of water-only parameters for OAMe–G2 cavity waters against bulk and against water within 10 Å of the portal, bridging waters against bulk, and T4L L99A cavity and buried waters against protein-surface water. (b) OAMe–G2: ROC area for identifying the waters near the portal that enter the cavity within the next ten saved states; single features evaluated on all paths, logistic models held out by path. (c) T4L L99A: waters within 8 Å of the cavity flagged as buried per frame by rules fitted on buried versus surface water of the other replica pair (mean of both directions), in the dry and wet ensembles; dashed, the three cavity waters of the wet ensemble.

Data. The unbinding trajectories belong to separate path-sampling campaigns and are not archived. The archive (`coordinates/applications/`) holds, per water and analyzed frame, every order parameter, the geometric region label, the quantities that define it (position relative to the host axis and portal, burial, distances to the ligand) and the split columns (`path`, `replica`), as `oame_features.npz` and `t4l_features.npz`. `host_guest.py analyze` and `confined_water.py analyze` reproduce every number and Fig. S1 from these files alone; we checked that the regenerated tables are byte-identical to the published ones.